

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: CSA1610 **COURSE CODE:** Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO3: Evaluate the effectiveness of classification techniques on real-time data for accurate prediction, enabling informed decision-making. (BL3,BL4,BL5)
Assessment Tool Weightage: 10%

QUESTIONS: 30

Total Marks:30

Annexure A

MCQ TEST

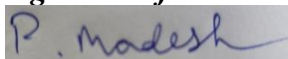
<i>Criteria</i>	Conceptual Understanding (3)	Accuracy of Answers (2.5)	Application of Knowledge (2)	Time Management (1.5)	Question Interpretation (1)	<i>Total(10)</i>
<i>Weightage</i>	30%	25%	20%	15%	10%	<i>100%</i>
<i>Q</i>						

Code of Conduct:

I _P. Madesh_192365007_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Conceptual Understanding	30%	Demonstrates thorough understanding; answers all questions accurately	Shows good understanding with few errors	Partial understanding; several incorrect responses	Lacks understanding; majority of answers incorrect
Accuracy of Answers	25%	All answers are correct	Most answers are correct with minor mistakes	Some correct answers; frequent errors	Mostly incorrect answers
Application of Knowledge	20%	Correctly applies concepts to problem-based questions	Applies concepts with minor errors	Limited ability to apply concepts	Unable to apply concepts
Time Management	15%	Completes quiz well within time efficiently	Completes on time with minor delays	Requires extra time or rushes through questions	Unable to complete within given time
Question Interpretation	10%	Interprets all questions correctly and responds appropriately	Misinterprets a few questions	Often misinterprets questions	Unable to understand questions

ANNEXURE A

MCQ Test:

1. Consider a simple linear regression model with one independent variable. Which of the following is true? (CO3)

Option_a: The regression line always passes through the mean of X and Y.

Option_b: The slope is always positive.

Option_c: The intercept is always positive.

Option_d: The correlation coefficient is always 1.

2. Which of the following statements is correct about Decision Trees? (CO3)

Option_a: Decision trees can only handle numerical data.

Option_b: Decision trees are always prone to overfitting.

Option_c: Decision trees can be used for classification and regression.

Option_d: None of the above.

3. Which of the following methods can be used for pruning in decision trees? (CO3)

Option_a: Pre-pruning

Option_b: Post-pruning

Option_c: Both (a) and (b)

Option_d: None of the above.

4. Which of the following is a disadvantage of Decision Trees? (CO3)

Option_a: Prone to overfitting.

Option_b: Requires large datasets.

Option_c: Cannot handle categorical data.

Option_d: Always has a high accuracy.

5. What does Bayes' Theorem primarily help with in probability and statistics? (CO3)

Option_a: Determining prior probabilities only

Option_b: Updating probabilities based on new evidence

Option_c: Computing standard deviation of a dataset

Option_d: Finding mean and median in distributions

6. In Bayes' Theorem, what does the term $P(A|B)$ represent? (CO3)

Option_a: The probability of event B given event A has occurred

Option_b: The probability of event A given event B has occurred

Option_c: The joint probability of A and B

Option_d: The marginal probability of B

7. A factory has two machines, A and B. Machine A produces 60% of the products, and machine B produces 40%. If 2% of A's products and 3% of B's products are defective, what is the probability that a randomly selected defective product came from Machine B? (CO3)

Option_a: 0.40

Option_b: 0.50

Option_c: 0.60

Option_d: 0.75

8. Which of the following is not required to apply Bayes' Theorem? (CO3)

Option_a: Prior probability of the event

Option_b: Conditional probability of the evidence

Option_c: Total number of possible outcomes

Option_d: Probability of the evidence

9. A doctor uses a test to diagnose a disease that affects 1% of a population. The test correctly identifies the disease 95% of the time but also gives a false positive in 5% of cases. If a patient tests positive, what is the probability they actually have the disease? (CO3)

Option_a: 0.16

Option_b: 0.34

Option_c: 0.95

Option_d: 0.50

10. Bayes' Theorem is most useful when dealing with which type of probability? (CO3)

Option_a: Independent events

Option_b: Conditional probability

Option_c: Mutually exclusive events

Option_d: Uniform probability distribution

11. If a hypothesis HHH has a prior probability of 0.3 and new evidence EEE has a likelihood of 0.8 given HHH, how does Bayes' Theorem adjust the probability of HHH after seeing EEE? (CO3)

Option_a: The probability of HHH increases

Option_b: The probability of HHH decreases

Option_c: The probability of HHH remains unchanged

Option_d: The probability of HHH is completely unknown

12. A bag contains 5 red and 7 blue balls. A ball is drawn randomly and found to be red. What is the probability that the next ball drawn is also red (assuming the first ball is not replaced)? (CO3)

Option_a: 5/12

Option_b: 4/11

Option_c: 5/11

Option_d: 6/12

13. Which field heavily relies on Bayes' Theorem for updating beliefs based on new data? (CO3)

Question 230: END

Option_a: Machine Learning

Option_b: Geometry

Option_c: Classical Physics

Option_d: Cryptography

14. Why is backpropagation crucial in training deep neural networks? (CO3)

Option_a: It prevents overfitting by pruning unnecessary weights.

Option_b: It updates weights using the gradient of the loss function.

Option_c: It ensures each layer has the same activation values.

Option_d: It removes redundant layers to optimize the model.

15. During backpropagation, what happens when the learning rate is too high? (CO3)

Option_a: The model converges faster with high accuracy.

Option_b: The model may overshoot the optimal weight values and fail to converge.

Option_c: The model will always get stuck in local minima.

Option_d: The number of hidden layers increases automatically.

16. What is the main advantage of rule-based induction models? (CO3)

Question 247: END

Option_a: They provide an interpretable set of rules for decision-making.

Option_b: They work well with both numerical and categorical data.

Option_c: They can generalize well without requiring large datasets.

Option_d: Both (a) and (b)

17. Which of the following is an example of a rule-based induction algorithm? (CO3)

Question 248: END

Option_a: Decision Tree

Option_b: Naïve Bayes

Option_c: Support Vector Machine

Option_d: K-Nearest Neighbors

18. In rule-based induction, what happens when the rules are too specific to the training data?

Option_a: The model may suffer from overfitting.

Option_b: The model will generalize better.

Option_c: The rules will be shorter and more efficient.

Option_d: The accuracy will always increase.

19. Which approach is commonly used to simplify rule-based induction models? (CO3)

Question 250: END

Option_a: Pruning redundant or less significant rules

Option_b: Adding more features to the dataset

Option_c: Increasing the depth of the decision tree

Option_d: Using more training samples

20. What is the primary purpose of a confusion matrix in evaluating a classifier? (CO3)

Option_a: To visualize the performance of a classifier by showing correct and incorrect predictions

Option_b: To calculate the computational complexity of the classifier

Option_c: To determine the training time of the classifier

Option_d: To identify the features used by the classifier

21. Which metric is used to measure the proportion of correctly classified instances out of the total instances? (CO3)

Question 252: END

Option_a: Precision

Option_b: Recall

Option_c: Accuracy

Option_d: F1-Score

22. In a binary classification problem, what does the True Positive (TP) value represent? (CO3)

Question 253: END

Option_a: The number of negative instances correctly classified as negative

Option_b: The number of positive instances correctly classified as positive

Option_c: The number of positive instances incorrectly classified as negative

Option_d: The number of negative instances incorrectly classified as positive

23. What is the formula for calculating precision in a classification problem? (CO3)

Option_a: Precision = $TP / (TP + FP)$

Option_b: Precision = $TP / (TP + FN)$

Option_c: Precision = $(TP + TN) / (TP + TN + FP + FN)$

Option_d: Precision = $TP / (FP + FN)$

24. Which of the following metrics is most useful when the classes are imbalanced? (CO3)

Option_a: Accuracy

Option_b: F1-Score

Option_c: Recall

Option_d: Specificity

25. What does a high recall value indicate in a classification model? (CO3)

Question 256: END

Option_a: The model has a low number of false positives

Option_b: The model has a low number of false negatives

Option_c: The model has a high number of true negatives

Option_d: The model has a high number of false positives

Correct_option: The model has a low number of false negatives

26. Which of the following is an example of a rule-based induction algorithm? (CO3)

Option_a: Decision Tree

Option_b: Naïve Bayes

Option_c: Support Vector Machine

Option_d: K-Nearest Neighbor

27. In rule-based induction, what happens when the rules are too specific to the training data? (CO3)

Option_a: The model may suffer from overfitting.

Option_b: The model will generalize better.

Option_c: The rules will be shorter and more efficient.

Option_d: The accuracy will always increase.

28. Which approach is commonly used to simplify rule-based induction models? (CO3)

Option_a: Pruning redundant or less significant rules

Option_b: Adding more features to the dataset

Option_c: Increasing the depth of the decision tree

Option_d: Using more training samples

29. What is the primary purpose of a confusion matrix in evaluating a classifier? (CO3)

Option_a: To visualize the performance of a classifier by showing correct and incorrect predictions

Option_b: To calculate the computational complexity of the classifier

Option_c: To determine the training time of the classifier

Option_d: To identify the features used by the classifier

30. Which metric is used to measure the proportion of correctly classified instances out of the total instances? (CO3)

Question 252: END

Option_a: Precision

Option_b: Recall

Option_c: Accuracy

Option_d: F1-Score

CO3 - AT3

85
100

1. A) The regression line always passes through the mean of x .
2. C) Decision trees can be used for classification and regression.
3. C) Both (a) and (b) - pre pruning and post-pruning.
4. A) prone to overfitting.
5. B) updating probability based on new evidence
6. B) probability of event A given event B has occurred
7. D) 0.75
8. C) Total number of possible outcomes.
9. A) 0.16
10. B) conditional probability.
11. A) The probability of H included
12. B) 4/11
13. B) machine learning.
14. B) It updates weights using the gradient of the loss function.
15. B) The model may overshoot the optimal weight values and fail to converge.

16. D) both (a) and (b)
17. A) Decision tree
18. A) The model may suffer from overfitting.
19. A) pruning redundant or less significant rules.
20. A) visualize classification performance by showing correct and incorrect predictions.
21. C) accuracy.
22. B) positive instances correctly classified as positive.
23. A) $\text{precision} = TP / (TP + FP)$
24. D) F1-Score.
25. B) The model has a low number of false negatives.
26. A) Decision tree
27. A) The model may suffer from overfitting.
28. A) pruning redundant or less significant rules.
29. A) visualize classifier performance by showing correct and incorrect predictions.
30. C) Accuracy.